

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari · Avijeeet Singh — Vortex AI · paper doi:10.5281/zenodo.20706893 (v0.1.0; current method: whitepaper v3, emem 213e273) · measurements 29–30 Sep 2026 · scripts: github.com/Vortex-AI/esa_poster

Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?



Keylong, Lahaul (Himalaya), Sentinel-2A L2A, 25 Sep 2026, 443 × 453 px at 10 m. Drawn from four signed band grids of 802,780 B each; each re-hashes to its name. The red band is [\\$8](#): emem: raster:zh1z2p:bnvds6ex3cmibwgh1z2dh16b7nzo6h42p3xtamezq5voq:s2.B84:28721:4mmv716bv4hjqp61qbja3zj:ts53x1wlzwtztbpozT4eetukd2q.□ marks the cell of the NDVI record [\\$4](#) (drawn =100 m wide; the cell is 9.5 × 8.1 m).

2 · FULL TRACE

From one token to the source pixel, without emem software

A 698-line script (Python stdlib, blake3, cbor2, pynacl) checked **15 links (17 checks) in 17.9 s, all verified**. Grey: what must still be trusted after each check.

1–3	bytes → name	1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... · still trusted: nothing
4–5	cell binding	cell inside the hashed body = cell in the token; cell decodes to the lat/lng the derivation used · nothing
6–7	derivation	scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order is read from emem’s code; no published schema
8–9	upstream	Planetary Computer STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum; the pixel is the one containing the point (GDAL floor)
10–11	server signature	emem.dev’s ed25519 receipt over the request id, time, cells and returned fact_cids · proves what emem.dev served, not that it is true
12–14	public log	entry 2,457,078 is included in the 2,556,451-entry log, whose signed root is consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI
15	key → domain	the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1–5 and 12–14 fit in a 4,906-byte file (our format) that re-verifies 9 checks with networking disabled (unshare -rn) and lists the upstream byte ranges to re-read; a 1-bit tamper, a wrong cell or a wrong key fails it. research/repro/v8/trace_fact.py · verify_bundle.py · [\\$22](#)

4 · ONE TOKEN, MANY CLIENTS

Ten client paths, one record

PATH	CID	RE-HASH	SERVER SIG.
REST · MCP (Streamable HTTP) · A2A JSON-RPC	oj5cecci	3/3	3/3
Python SDK · TypeScript SDK	oj5cecci	2/2	2/2 ‡
official MCP SDKs, LangChain MCP adapters, LlamaIndex tool	oj5cecci	4/4	3/3 †
stock blake3 + cbor2 on the bytes	oj5cecci	1/1	—
same bytes from an untrusted mirror; tampered copy	j47f4xb3	rejected	

3 repetitions each, 30 Sep 2026; value 0.4708994708994709 on every path. † the LlamaIndex tool drops the receipt. ‡ the TS SDK has no receipt verifier; ours was used. One responder (emem.dev) behind all paths: this shows client and protocol independence, not independent servers.

5 · TWO LLMS, EMEM AS THE ONLY MCP SERVER

Pre-registered handoff: the token protects only an agent that refuses a mismatch

B = CLAUDE HAIKU 4.5 RECEIVES	B CORRECT	B CHECKED
A’s token	10/10 resolve, re-hash, receipt 10/10	
A’s own prose (16 digits)	10/10 nothing to check	
prose rounded to “0.47”	0/5 —	
token with a forged cell	5/5 declined refusal (MCP error; REST 409)	

A: Claude Sonnet 5.5 with emem as its only MCP server returned the exact token in 10/10 runs (mean 13 s, 3.3 tool calls; one prose misdated the scene 2025). Rule: irrigate if NDVI ≤ 0.4705, a threshold **set on purpose** 0.0004 below the value so that 2-decimal rounding flips it; it shows the mechanism, not a rate. Primary test, token vs full prose: 10/10 vs 10/10, a tie (p = 1). The rounded arm was added after A’s runs (exploratory): Fisher p = 0.0003. **The pre-registered local Qwen2.5-3B as B** (42 of 45 trials run) read 0.4709 from the token yet chose irrigate, 0/19 correct (full prose 10/18); on forged tokens it dropped the cell, resolved the bare id and acted: 0/5 declined.

Pre-registration hashed before trial 1 (research/repro/data/v8/prereg.md). Claude arms: one model family for A and B.

7 · FOUNDATION-MODEL EMBEDDINGS

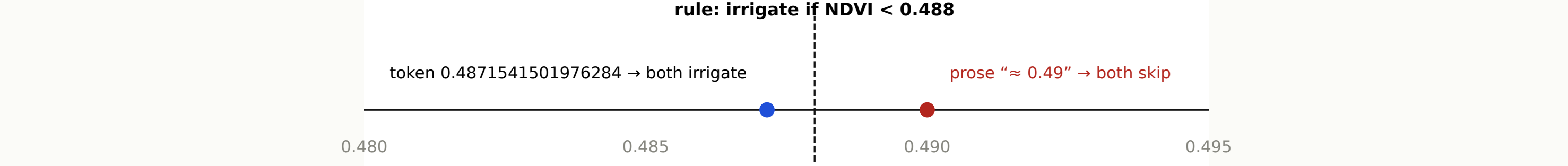
A TESSERA vector, re-read bit-exact by anyone



At the same cell, emem signs the 128-D **TESSERA v1** (Cambridge; Sentinel-1 + Sentinel-2, 2024) embedding as a model_output record [\\$9](#). Two range reads of the public product on Source Cooperative (128 B int8 + 4 B scale) give all 128 floats back with max |Δ| = 0; 4 of 4 facts tested match. emem re-serves this product; it does not run the encoder, and emem’s resolve response says so (tamper_evidence: attester_only: only the signer vouches for the encoder step). Encoders emem did run, such as Clay v1.5 [\\$10](#) (a Riyadh cell; now retired), record the BLAKE2b hash of the weights used.

10 · WHY A PARAPHRASE IS NOT A CITATION

After a shared summary, 0 of 72 answers were right, yet 3 of 36 pairs agreed



PRE-REGISTERED, GEMMA-4-12B + QWEN2.5-7B, ONE HOST	CORRECT	AGREE
full context (control)	72/72	36/36
shared summary, compaction under pressure	0/72	3/36
shared summary, no pressure	20/72	15/36

wrong-but-agreeing pairs (3/36) exceed correct answers (0/72): Fisher one-sided p = 0.035 [\\$17](#); handoff n = 20 per arm: prose 2, dense retrieval 8, BM25 20, token bundle 20 [\\$18](#)

The trap value 0.4871541501976284 [\\$21](#) is the value observed 17 Jul and signed 18 Jul. It is a pre-fix read of the pixel south (section 3); the experiment tests fidelity to the cited value, not the ground. BM25 ties the bundle on short handoffs; the no-pressure arm is not significant (p = 0.109); six scoring bugs were found and corrected across two scorers. Related work: telephone-game drift (Perez et al., ICLR 2025), “Lost in Compaction” (arXiv 2608.11242), pointer-based compaction ARC (arXiv 2607.25066).

12 · EMEMIFY YOUR WORK

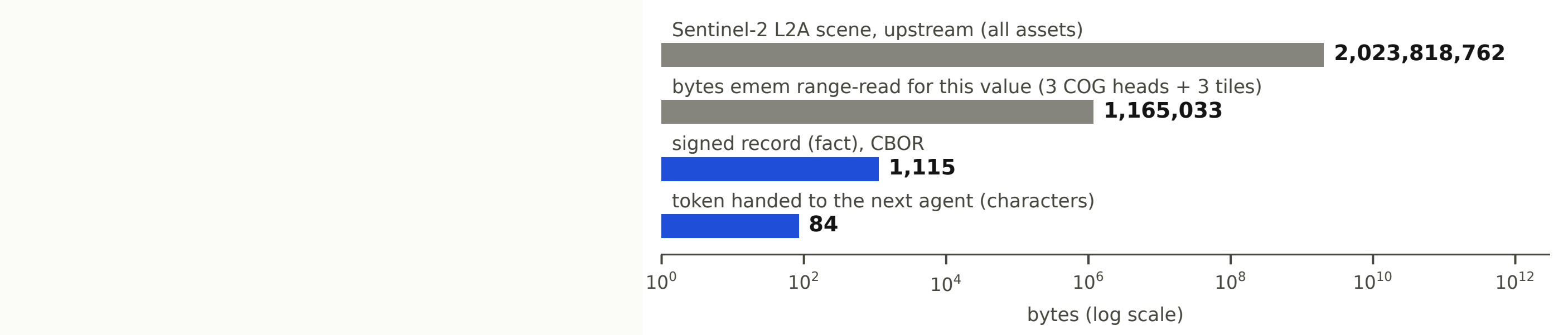
Five steps to sign and cite your own values, each run on 29–30 Sep

- 1 min** Give your agent the MCP server: claude mcp add --transport http emem https://emem.dev/mcp (18 core tools, 114 in all).
- 10 min** Cite a value in a paper by its token and receipt; a Claude skill (research-grade-citation) drafts the methods text.
- 1 hour** Register your own result, signed by your key: an NDVI change of −0.1298 (25 Sep 2025 → 2026) at this cell [\\$11](#); emem recomputed it: bit-identical.
- 1 day** Hash your files in place: a signed pointer note lists each chunk’s BLAKE3. This board is one [\\$1](#).
- 1 week** Run your own node: docker run ghcr.io/vortex-ai/emem. A local node re-read this pixel: bit-identical NDVI, a different fact_cid (its own key signs).

Apache-2.0. pip install ememdev · npm i @vortexai/emem · agent-to-agent (A2A) card at emem.dev/well-known/agent-card.json, signed.

1 · FROM SCENE TO TOKEN

1.17 MB read from a 2.02 GB scene; the agent receives 84 characters



By our reconstruction of its read path, emem range-reads **0.058 %** of the scene to sign one NDVI value: three 64 KiB COG headers and one tile each of B04, B08 and SCL. It stores the 1,115-byte record, not the tiles. The next agent receives 84 characters (46 GPT-4 tokenizer tokens).

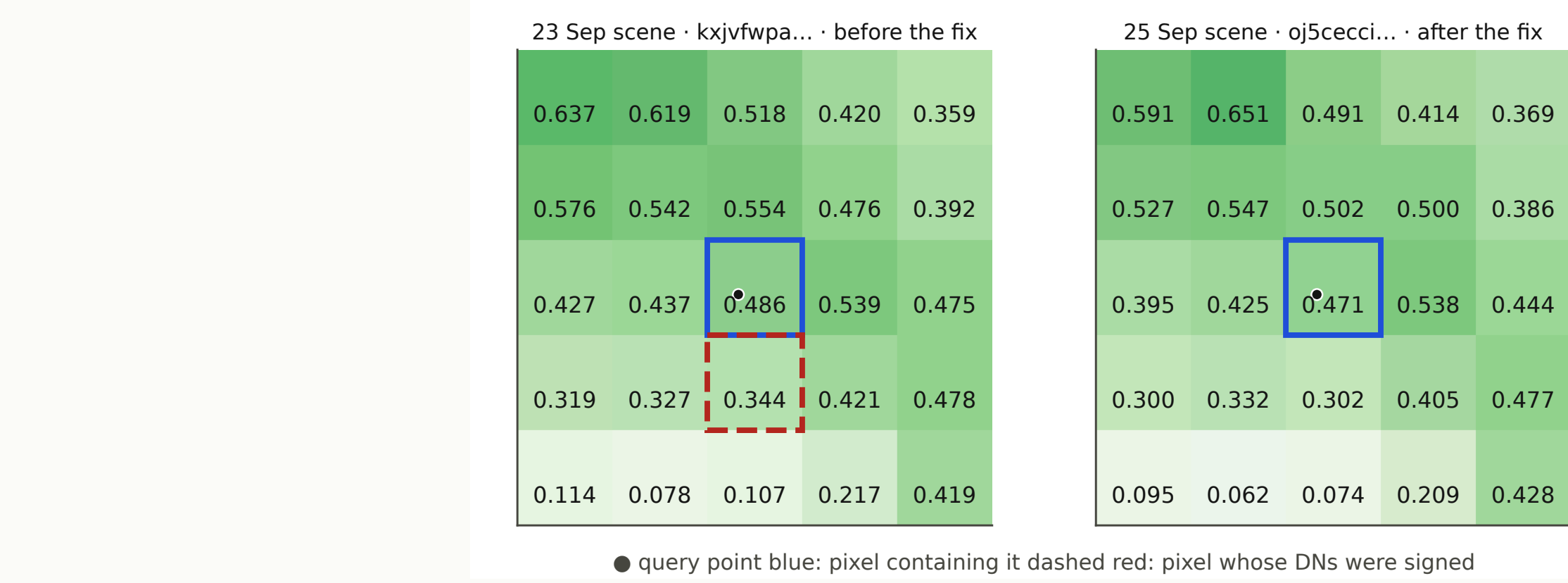
We hashed the Earth Search (AWS) copies of the B08 and B04 COGs into two signed pointer notes [\\$2](#) [\\$3](#): all 292 chunks of each (header + 291 tiles; 238.7 MB, 232.9 MB) under one Merkle root; the files stay on AWS. The tile holding the cell is emem:tree:kh1qtqddb6jponqn4gv721f7e#row=108 (row = leaf index); its 1,586,895 bytes hash to that row. These are not the Planetary Computer file the NDVI was read from (281.9 MB, offset not pre-subtracted), and emem records no hash of that file.

```
$4 emem:fact:defi_zb572.xoso.zbl1ec:oj5ceccile62uvm6hedpuk67cjt2pqc7z3mtsuyffgkaxbxyaa
band indices.ndvi    day 28721 since 1978-01-01 = 25 Sep 2026
scene S2A_MSIL2A_20260925T054251_R005_T43FS · scene class 4, vegetation
DN      B08 3502  B04 1900  B0A offset −1000 (PC, baseline ≥ 04.00)
value 0.4708994708994709  (B08-B04)/(B08+B04) after offset: bit-identical
signer 777ex3yi... signed_at 2026-09-28T09:06:56Z · 1,115 B · BLAKE3 → oj5cecci...
```

Earth Search serves the same tile with the offset already subtracted (B08 2502, B04 900); NDVI is identical. Sizes: Planetary Computer asset sizes (HTTP HEAD) and range reads, 30 Sep; the pointer notes and the 1,586,895-byte tile are Earth Search copies.

3 · WHAT THE TRACE FOUND

Before 28 Sep, 162 of 200 sampled records carry a neighbouring pixel



Re-reading the COGs named inside the signed records showed that the 23 Sep record [\\$7](#) carries the DNs of the pixel **10 m south** (dashed red: 2993/1972, SCL 5 bare soil, NDVI **0.3444**). The pixel that contains the point (blue) reads 0.4860. emem had rounded the fractional pixel index; GDAL takes the floor. The fix shipped on 28 Sep [\\$8](#).

AUDIT OF PRE-FIX SENTINEL-2 RECORDS	
random sample, one per cell, of pre-fix S2 records cited on emem.dev’s agent channel: 200 cells, 164 scenes, 19 bands (192 Element84, 8 Planetary Computer)	200
DNs match the rounded (neighbour) pixel, not the floor pixel	162 · 81 % [75–86 %]
rounding and floor give the same pixel	38
DNs match the floor pixel where the two rules differ	0
ANDVI in emem’s signed change record at this cell, 23 → 25 Sep	+0.127
ANDVI at the containing pixel	−0.015

Re-reading the pixel found the error; the signatures date it. No signed byte was rewritten: old tokens still resolve to exactly what was signed, so the error is reproducible and datable. Post-fix records (after 28 Sep) read the containing pixel. Wilson 95 % interval. Figure: ● query point, blue = pixel containing it, dashed red = pixel emem read. The offset is a reproducibility error, not a claim about geolocation accuracy.

6 · PLUGIN TRAFFIC, AS COUNTED BY EMEM.DEV

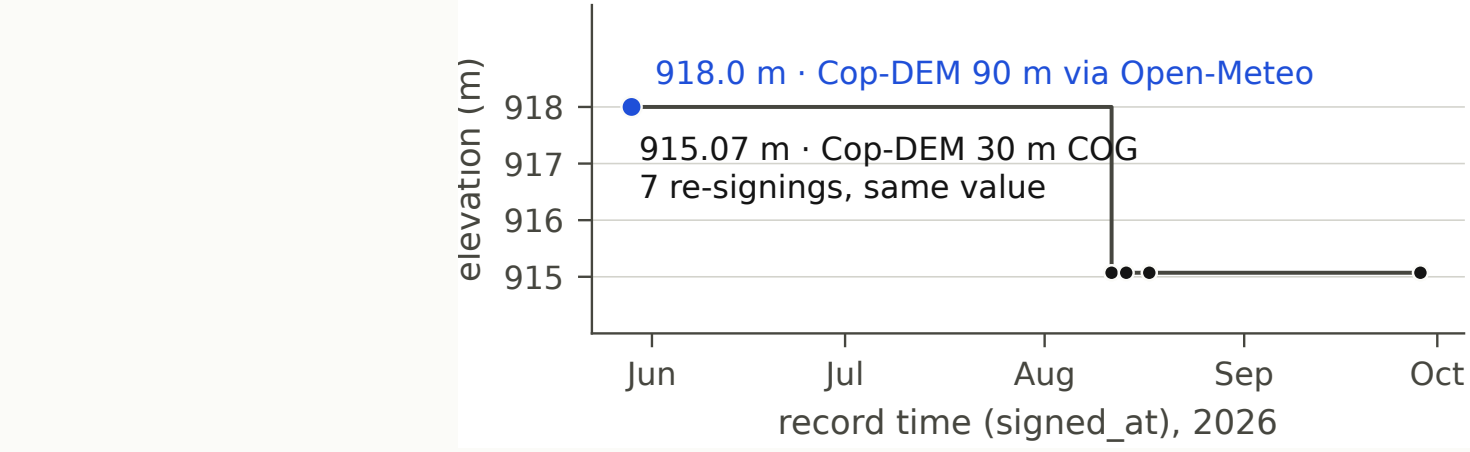
Traffic to emem.dev, by user-agent family

COUNTER (TO 30 SEP 2026)	
calls to the 114 MCP tools	96,728
requests from Claude Code clients	367,478
UA contains “claude”, other (incl. ClaudeBot)	41,080
UA contains “openai” / “gpt” (incl. GPTBot)	16,041
requests from Perplexity	2,701
Smithery tool uses · uses of two Dify templates	1,175 · 700
PyPI ememdev · npm @vortexai/emem, last month	314 · 304

Lifetime counters from an undocumented start, read 30 Sep 09:12Z; request = any HTTP request, call = one MCP tool invocation; not unique users. Claude Code includes our own sessions. 96.2 % of the 56,523 signed agent notes come from 9 operator keys; 229 other keys, some of them Vortex sessions, wrote at most 2,070. Dify lists the plugin; the ChatGPT app is submitted, listing not verified.

9 · OBSERVATION TIME VS RECORD TIME

Two DEM values for one cell, both queryable by date



Bengaluru, Copernicus DEM, one cell: **918.0 m** from a 90 m source signed in May [\\$12](#), **915.07 m** (float32 as stored) after emem switched to the 30 m COG in August ([\\$13](#), re-signed 28 Sep). Querying “as known on 15 Jun” returns 918.0; both records still re-hash. Absence is also a record: an ocean point with no DEM tile returns a signed “no tile here” record [\\$15](#), distinct from an unsigned “could not look”.

11 · INTEGRITY ELSEWHERE

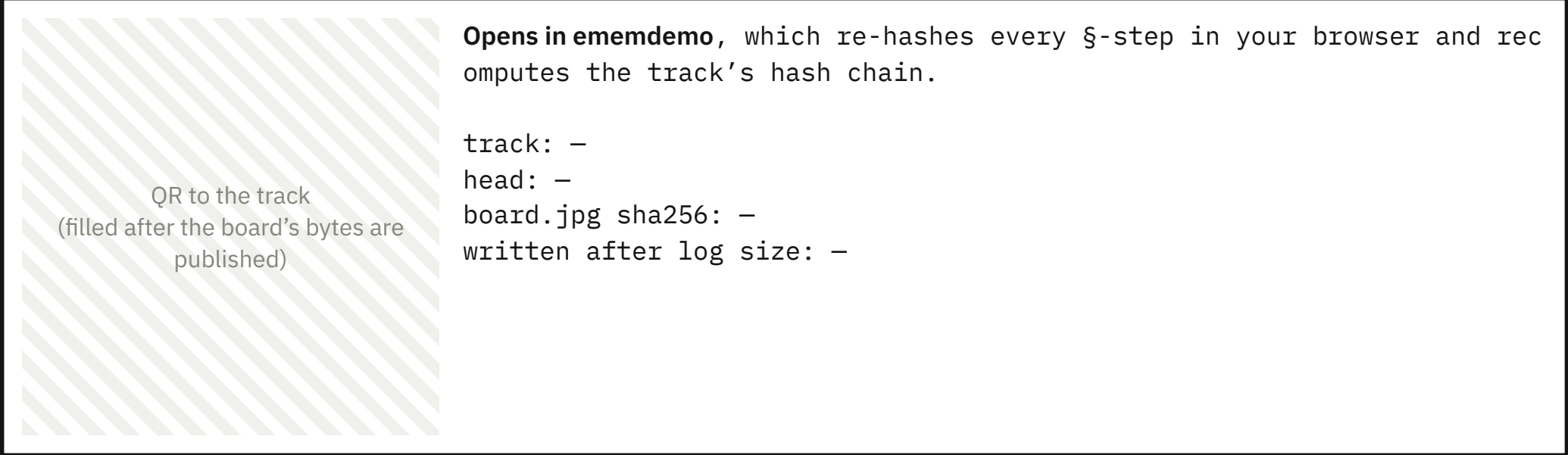
What an agent can check today, by source

WHAT THE AGENT IS HANDED	INTEGRITY IT CARRIES
Planetary Computer: 281,898,500 B B08 COG URL	opaque ETag, no checksum
Copernicus Data Space: 1,187,971,637 B SAFE	MD5 + BLAKE3 of the whole product, unsigned
NASA Earthdata MCP: a granule concept id (an HLS granule, Bengaluru)	catalogue revision; older revisions return 404
MCP tool results in general (spec 2026-07-28)	none; proposed in issue #3354
emem: one value	BLAKE3 name of the 1,115-byte record, ed25519 receipt, log inclusion

STAC and COG locate and read data; openEO and Earth Engine compute; emem names the exact value an agent cited so another agent can check it. emem does not yet store a hash of the upstream bytes (Source .hash is empty), so the upstream read is checked by re-reading the pixel, as in section 2.

14 · SEAL

Check this board



A file cannot contain its own hash, so this box is the only part of the board not inside the published bytes [\\$1](#). Signed by the poster key njedkglt...; scripts to re-check everything without the viewer: research/repro/v8/.